

NTA CUET PAPER (29TH MAY 2024)

Instructions

Attempt only 40 questions out of the given 50 questions. Each question carries 5 marks. One mark will be deducted for a wrong answer.

Full Marks: 200

(Time: 60 Minutes)

1. The kinetic energy of an electron in the ground level of a hydrogen atom is K units. The values of its potential energy and total energy respectively are _____.

(a) $-2K$; $-K$
(b) $+2K$; $-K$
(c) $-K$; $+2K$
(d) $+K$; $+2K$

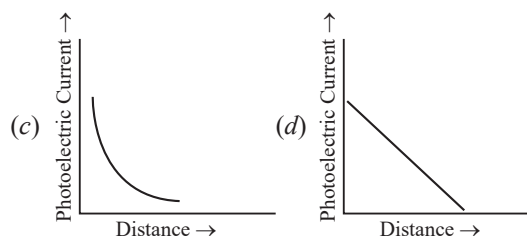
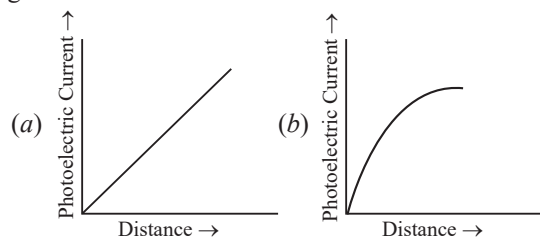
2. Two nuclei have mass numbers A and B respectively. The density ratio of the nuclei is _____.

Fill in the blank with the correct answer from the options given below.

(a) $A : B$
(b) $\sqrt{A} : \sqrt{B}$
(c) $A^2 : B^2$
(d) $1 : 1$

3. A point source causing photoelectric emission from a metallic plate is moved away from the plate. The variation of photoelectric current with distance from the source is correctly represented by the graph _____.

Fill in the blank with the correct answer from the options given below.



4. A proton accelerated through a potential difference V has a de Broglie wavelength λ . On doubling the accelerating potential, de Broglie wavelength of the proton _____.

Fill in the blank with the correct answer from the options given below.

(a) remains unchanged
(b) becomes double
(c) becomes four times
(d) decreases

5. The shortest wavelengths emitted in hydrogen spectrum corresponding to different spectral series are as under:

A. Pfund series B. Balmer series
C. Brackett series D. Lyman series

The wavelengths arranged correctly in decreasing order are _____.

Fill in the blank with the correct answer from the options given below.

(a) (A), (B), (C), (D) (b) (A), (C), (B), (D)
(c) (B), (A), (D), (C) (d) (A), (C), (D), (B)

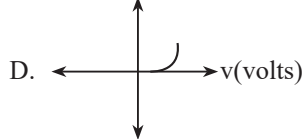
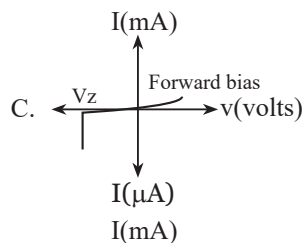
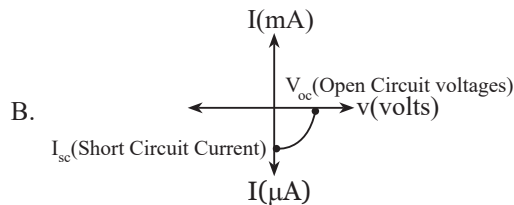
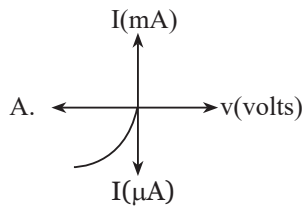
6. Silicon can be doped using one of the following elements as dopant:

A. Arsenic
B. Indium
C. Phosphorus
D. Boron

To get n-type semiconductor, the dopants that can be used are _____.

Fill in the blank with the correct answer from the options given below.

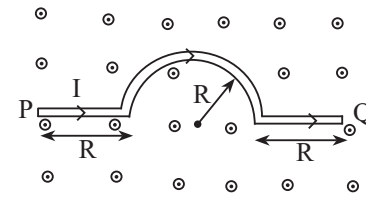
- (a) (A) and (C) only
(b) (B) and (C) only
(c) (A), (B), (C) and (D)
(d) (C) and (D) only
7. Given below are V versus I graphs for different types of p-n junction diodes marked A, B, C, and D.



The correct sequence of graphs corresponding to forward biased p-n junction, Zener diode, photo diode, and solar cell in order is _____.

Fill in the blank with the correct answer from the options given below.

- (a) (D), (C), (A), (B)
(b) (A), (C), (B), (D)
(c) (B), (A), (D), (C)
(d) (C), (B), (D), (A)
8. A wire carrying current I , bent as shown in the figure, is placed in a uniform field B that emerges normally out of the plane of the figure. The force on this wire is _____.



Fill in the blank with the correct answer from the options given below.

- (a) $4BIR$, directed vertically downward
(b) $3BIR$, directed vertically upward
(c) $BI(2R + \pi R)$, vertically downward
(d) $2\pi BIR$, from P to Q
9. The refractive index of the material of an equilateral prism is $\sqrt{2}$. The angle of minimum deviation of that prism is _____.

Fill in the blank with the correct answer from the options given below.

- (a) 60°
(b) 75°
(c) 30°
(d) 90°
10. The transfer of integral number of _____ is one of the evidence of quantization of electric charge.

Fill in the blank with the correct answer from the options given below.

- (a) photons
(b) nuclei
(c) electrons
(d) neutrons
11. When a slab of insulating material 4 mm thick is introduced between the plates of a parallel plate capacitor of separation 4 mm, it is found that the distance between the plates has to be increased by 3.2 mm to restore the capacity to its original value. The dielectric constant of the material is _____.

Fill in the blank with the correct answer from the options given below.

- (a) 2
(b) 5
(c) 3
(d) 7
12. A copper ball of density 8.0 g/cc and 1 cm in diameter is immersed in oil of density 0.8 g/cc. The charge on the ball if it remains just suspended in oil in an electric field of intensity 600 N/C acting in the upward direction is _____.

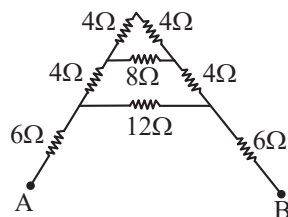
Fill in the blank with the correct answer from the options given below. (Take $g = 10 \text{ m/s}^2$)

- (a) $2 \times 10^{-6} \text{ C}$
(b) $2 \times 10^{-5} \text{ C}$
(c) $1 \times 10^{-5} \text{ C}$
(d) $1 \times 10^{-6} \text{ C}$
13. A metal wire is subjected to a constant potential difference. When the temperature of the metal wire increases, the drift velocity of the electron in it _____.

Fill in the blank with the correct answer from the options given below.

- (a) increases, thermal velocity of the electrons decreases
 (b) decreases, thermal velocity of the electrons decreases
 (c) increases, thermal velocity of the electrons increases
 (d) decreases, thermal velocity of the electrons increases

14. For the given mixed combination of resistors, calculate the total resistance between points A and B.



Choose the correct answer from the options given below.

- (a) 9 Ω (b) 18 Ω
 (c) 4 Ω (d) 14 Ω
15. A cell of emf 1.1 V and internal resistance 0.5 Ω is connected to a wire of resistance 0.5 Ω . Another cell of the same emf is now connected in series with the intention of increasing the current but the current in the wire remains the same. The internal resistance of the second cell is _____.
- Fill in the blank with the correct answer from the options given below.
- (a) 1 Ω (b) 2.5 Ω
 (c) 1.5 Ω (d) 2 Ω
16. P, Q, R, and S are four wires of resistances 3 Ω , 3 Ω , 3 Ω , and 4 Ω respectively. They are connected to form the four arms of a Wheatstone bridge circuit. The resistance with which S must be shunted in order that the bridge may be balanced is _____.

Fill in the blank with the correct answer from the options given below.

- Options:
 (a) 14 Ω (b) 12 Ω
 (c) 15 Ω (d) 7 Ω

17. Magnetic moment of a thin bar magnet is 'M'. If it is bent into a semicircular form, its new magnetic moment will be _____.

Fill in the blank with the correct answer from the options given below.

- (a) M/π (b) $M/2$
 (c) M (d) $2M/\pi$

18. Ferromagnetic material used in Transformers must have _____.

Fill in the blank with the correct answer from the options given below.

- (a) Low permeability and High Hysteresis loss

- (b) High permeability and Low Hysteresis loss
 (c) High permeability and High Hysteresis loss
 (d) Low permeability and Low Hysteresis loss

19. A conducting ring of radius r is placed in a varying magnetic field perpendicular to the plane of the ring. If the rate at which the magnetic field varies is x , the electric field intensity at any point of the ring is _____.

Fill in the blank with the correct answer from the options given below.

- (a) rx (b) $rx/2$
 (c) $2rx$ (d) $4\pi r/x$

20. A 50 Hz AC current of crest value 1 A flows through the primary of a transformer. If the mutual inductance between the primary and secondary be 0.5 H, the crest voltage induced in the secondary is _____.

Fill in the blank with the correct answer from the options given below.

- (a) 75 V (b) 150 V
 (c) 100 V (d) 200 V

21. A long solenoid of diameter 0.1 m has 2×10^4 turns per meter. At the center of the solenoid, a coil of 100 turns and radius 0.01 m is placed with its axis coinciding with the solenoid axis. The current in the solenoid reduces at a constant rate to 0 A from 4 A in 0.05 s. If the resistance of the coil is $10\pi^2 \Omega$, then the total charge flowing through the coil during this time is _____.

Fill in the blank with the correct answer from the options given below.

- (a) 16 μC (b) 32 μC
 (c) $16\pi \mu\text{C}$ (d) $32\pi \mu\text{C}$

22. Lower half of a convex lens is made opaque. Which of the following statements describes the image of the object placed in front of the lens?

Choose the correct answer from the options given below.

- A. No change in image
 B. Image will show only half of the object
 C. Intensity of image gets reduced
 (a) (A) only (b) (B) only
 (c) (C) only (d) (B) and (C) only

23. Two slits are made 0.1 mm apart and the screen is placed 2 m away. The fringe separation when a light of wavelength 500 nm is used is _____.

Fill in the blank with the correct answer from the options given below.

- (a) 1 cm (b) 0.15 cm
 (c) 1.5 cm (d) 0.1 cm

Therefore, the fringe separation is 1 cm.

24. For an astronomical telescope having an objective lens of focal length 10 m and an eyepiece lens of focal length 10 cm, the telescope's tube length and magnification respectively are _____.

Fill in the blank with the correct answer from the options given below.

- (a) 20 cm, 1 (b) 1000 cm, 1
(c) 1010 cm, 1 (d) 1010 cm, 100

25. According to Bohr's Model:

- A. The radius of the orbiting electron is directly proportional to 'n'.
B. The speed of the orbiting electron is directly proportional to '1/n'.
C. The magnitude of the total energy of the orbiting electron is directly proportional to '1/n²'.
D. The radius of the orbiting electron is directly proportional to 'n²'.

Choose the correct answer from the options given below.

- (a) (A), (B) and (C) only
(b) (A), (B) and (D) only
(c) (A), (B), (C) and (D)
(d) (B), (C) and (D) only

26. For a full wave rectifier, if the input frequency is 50 Hz, the output frequency will be _____.

Fill in the blank with the correct answer from the options given below.

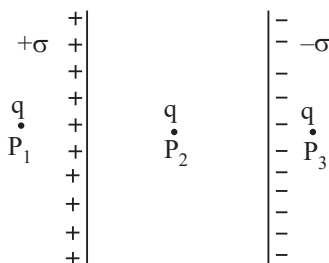
- (a) 50 Hz (b) 100 Hz
(c) 25 Hz (d) 0 Hz

27. For an electric dipole in a non-uniform electric field with dipole moment parallel to direction of the field, the force (F) and torque (τ) on the dipole respectively are _____.

Fill in the blank with the correct answer from the options given below.

- (a) $F = 0, \tau = 0$
(b) $F \neq 0, \tau = 0$
(c) $F = 0, \tau \neq 0$
(d) $F \neq 0, \tau \neq 0$

28. Two large plane parallel sheets shown in the figure have equal but opposite surface charge densities $+\sigma$ and $-\sigma$. A point charge q placed at points P₁, P₂, and P₃ experiences forces F₁, F₂, and F₃ respectively. Then:



Choose the correct answer from the options given below.

- (a) $F_1 = 0, F_2 = 0, F_3 = 0$
(b) $F_1 = 0, F_2 \neq 0, F_3 = 0$
(c) $F_1 \neq 0, F_2 \neq 0, F_3 \neq 0$
(d) $F_1 = 0, F_3 \neq 0, F_2 = 0$

29. Two charged metallic spheres with radii R_1 and R_2 are brought in contact and then separated. The ratio of final charges Q_1 and Q_2 on the two spheres respectively will be _____.

Fill in the blank with the correct answer from the options given below.

- (a) $Q_1/Q_2 = R_2/R_1$
(b) $Q_1/Q_2 < R_1/R_2$
(c) $Q_1/Q_2 > R_1/R_2$
(d) $Q_1/Q_2 = R_1/R_2$

30. Two charged particles, placed at a distance d apart in vacuum, exert a force F on each other. Now, each of the charges is doubled. To keep the force unchanged, the distance between the charges should be changed to _____.

Fill in the blank with the correct answer from the options given below.

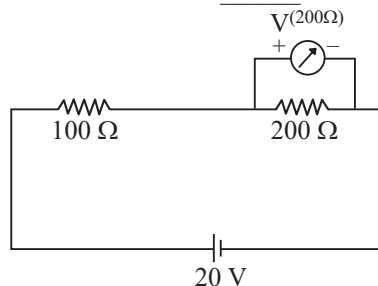
- (a) $4d$ (b) $2d$
(c) d (d) $d/2$

31. Two parallel plate capacitors of capacitances $2 \mu\text{F}$ and $3 \mu\text{F}$ are joined in series and the combination is connected to a battery of V volts. The values of potential across the two capacitors V_1 and V_2 and energy stored in the two capacitors U_1 and U_2 respectively are related as _____.

Fill in the blank with the correct answer from the options given below.

- (a) $V_1/V_2 = U_1/U_2 = 3/2$
(b) $V_1/V_2 = 2/3$ and $U_1/U_2 = 3/2$
(c) $V_1/V_2 = 3/2$ and $U_1/U_2 = 2/3$
(d) $V_1/V_2 = 2/3$ and $U_1/U_2 = 2/3$

32. Two resistances of 100Ω and 200Ω are connected in series across a 20 V battery as shown in the figure below. The reading in a 200Ω voltmeter connected across the 200Ω resistance is _____.



Fill in the blank with the correct answer from the options given below.

- (a) 4 V (b) $20/3 \text{ V}$
(c) 10 V (d) 16 V

33. The current through a $4/3 \Omega$ external resistance connected to a parallel combination of two cells of 2 V and 1 V emf and internal resistances of 1Ω and 2Ω respectively is _____.

Fill in the blank with the correct answer from the options given below.

- (a) 1 A (b) $2/3$ A
(c) $8/9$ A (d) $5/6$ A

34. A metallic wire of uniform area of cross-section has a resistance R , resistivity ρ , and power rating P at V volts. The wire is uniformly stretched to reduce the radius to half the original radius. The values of resistance, resistivity, and power rating at V volts are now denoted by R' , ρ' , and P' respectively. The corresponding values are correctly related as _____.

Fill in the blank with the correct answer from the options given below.

- (a) $\rho' = 2\rho$, $R' = 2R$, $P' = 2P$
(b) $\rho' = (1/2)\rho$, $R' = (1/2)R$, $P' = (1/2)P$
(c) $\rho' = \rho$, $R' = 16R$, $P' = (1/16)P$
(d) $\rho' = \rho$, $R' = (1/16)R$, $P' = 16P$

35. Three magnetic materials are listed below:

- A. Paramagnetics
B. Diamagnetics
C. Ferromagnetics

Choose the correct order of the materials in increasing order of magnetic susceptibility.

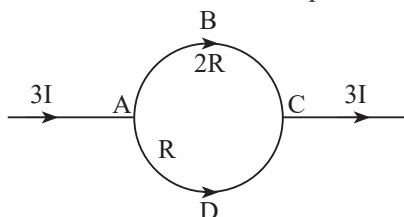
- (a) (A), (B), (C) (b) (C), (A), (B)
(c) (B), (A), (C) (d) (B), (C), (A)

36. Two infinitely long straight parallel conductors carrying currents I_1 and I_2 are held at a distance d apart in vacuum. The force F on a length L of one of the conductors due to the other is _____.

Fill in the blank with the correct answer from the options given below.

- (a) proportional to L but independent of $I_1 \times I_2$
(b) proportional to $I_1 \times I_2$ but independent of length L
(c) proportional to $I_1 \times I_2 \times L$
(d) proportional to $L / (I_1 \times I_2)$

37. In the circuit shown below, a current $3I$ enters at A. The semicircular parts ABC and ADC have equal radii ' r ' but resistances $2R$ and R respectively. The magnetic field at the center of the circular loop ABCD is _____.



Fill in the blank with the correct answer from the options given below.

- (a) $(\mu_0 I)/(4r)$ out of the plane
(b) $(\mu_0 I)/(4r)$ into the plane
(c) $(\mu_0 3I)/(4r)$ out of the plane
(d) $(\mu_0 3I)/(4r)$ into the plane

38. A square loop with each side 1 cm, carrying a current of 10 A, is placed in a magnetic field of 0.2 T. The direction

of the magnetic field is parallel to the plane of the loop. The torque experienced by the loop is _____.

Fill in the blank with the correct answer from the options given below.

- (a) zero
(b) 2×10^{-4} Nm
(c) 2×10^{-2} Nm
(d) 2 Nm

39. In an AC circuit, the current leads the voltage by $\pi/2$. The circuit is _____.

Fill in the blank with the correct answer from the options given below.

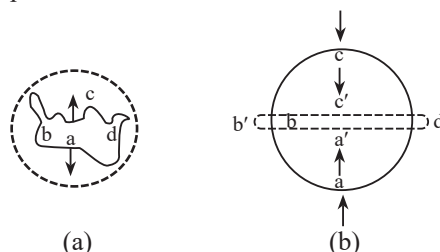
- (a) purely resistive
(b) should have circuit elements with resistance equal to reactance
(c) purely inductive
(d) purely capacitive

40. In a pair of adjacent coils, for a change of current in one of the coils from 0 A to 10 A in 0.25 s, the magnetic flux in the adjacent coil changes by 15 Wb. The mutual inductance of the coils is _____.

Fill in the blank with the correct answer from the options given below.

- (a) 120 H (b) 12 H
(c) 1.5 H (d) 0.75 H

41. A wire of irregular shape in figure (a) and a circular loop of wire in figure (b) are placed in different uniform magnetic fields as shown in the figures below. In figure (a), the magnetic field is perpendicular into the plane. In figure (b), the magnetic field is perpendicular out of the plane.

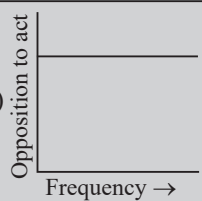
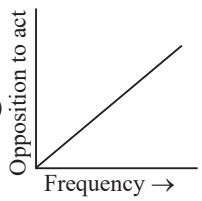
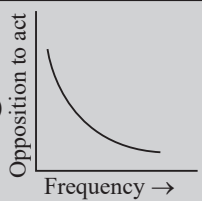
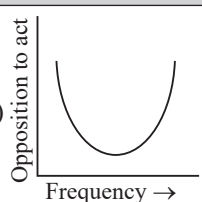


The wire in figure (a) is turning into a circular loop and that in figure (b) into a narrow straight wire. The direction of induced current will be _____.

Fill in the blank with the correct answer from the options given below.

- (a) clockwise in both (a) and (b)
(b) anticlockwise in both (a) and (b)
(c) clockwise in (a) and anticlockwise in (b)
(d) anticlockwise in (a) and clockwise in (b)

42. Match List-I has four graphs showing variation of opposition to flow of AC versus frequency with circuit characteristic in List-II.

List-I	List-II
(A) 	(I) Impedance
(B) 	(II) Capacitive reactance
(C) 	(III) Inductive reactance
(D) 	(IV) Resistance

Choose the correct answer from the options given below:

- (a) (A) – (I), (B) – (III), (C) – (II), (D) – (IV)
 (b) (A) – (IV), (B) – (III), (C) – (II), (D) – (I)
 (c) (A) – (I), (B) – (II), (C) – (IV), (D) – (III)
 (d) (A) – (III), (B) – (IV), (C) – (I), (D) – (II)

43. In an electromagnetic wave, the ratio of energy densities of electric and magnetic fields is _____.

Fill in the blank with the correct answer from the options given below.

- (a) 1 : 1 (b) 1 : c
 (c) c : 1 (d) 1 : c²

44. Of the following, the correct arrangement of electromagnetic spectrum in decreasing order of wavelength is _____.

Fill in the blank with the correct answer from the options given below.

- (a) Radio waves, X-rays, Infrared waves, microwaves, visible waves
 (b) Infrared waves, microwaves, Radio waves, X-rays, visible waves
 (c) Radio waves, microwaves, Infrared waves, visible waves, X-rays
 (d) X-rays, visible waves, Infrared waves, microwaves, Radio waves

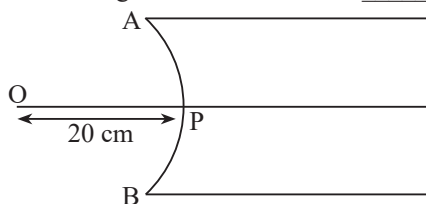
45. Match Electromagnetic waves listed in List-I with Production method/device in List-II.

List-I	List-II
(A) Microwaves	(I) LC oscillator
(B) Infrared	(II) Magnetron
(C) X-rays	(III) Vibration of atoms/molecules
(D) Radio waves	(IV) Bombarding large atomic number metal target with moving electrons

Choose the correct answer from the options given below:

- (a) (A) – (I), (B) – (II), (C) – (III), (D) – (IV)
 (b) (A) – (II), (B) – (III), (C) – (IV), (D) – (I)
 (c) (A) – (II), (B) – (I), (C) – (IV), (D) – (III)
 (d) (A) – (III), (B) – (IV), (C) – (I), (D) – (II)

46. In the figure given below, APB is a curved surface of radius of curvature 10 cm separating air and a transparent material ($\mu = 4/3$). A point object O is placed in air on the principal axis of the surface 20 cm from P. The distance of the image of O from P will be _____.



Fill in the blank with the correct answer from the options given below:

- (a) 16 cm left of P in air
 (b) 16 cm right of P in water
 (c) 20 cm right of P in water
 (d) 20 cm left of P in air

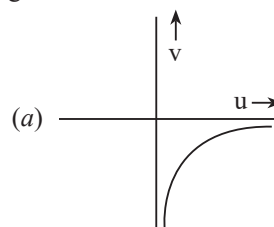
47. For fixed values of radii of curvature of lens, power of the lens will be _____.

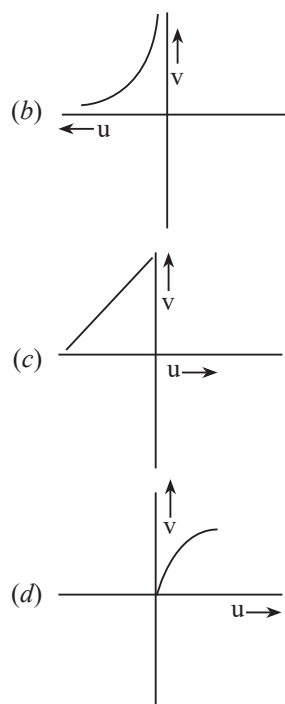
Fill in the blank with the correct answer from the options given below:

- (a) $P \propto (\mu - 1)$ (b) $P \propto \mu^2$
 (c) $P \propto 1/\mu$ (d) $P \propto \mu^{-2}$

48. The graph correctly representing the variation of image distance 'v' for a convex lens of focal length 'f' versus object distance 'u' is _____.

Fill in the blank with the correct answer from the options given below:





49. Using light from a monochromatic source to study diffraction in a single slit of width 0.1 mm, the linear width of central maxima is measured to be 5 mm on a screen held 50 cm away. The wavelength of light used is _____.

Fill in the blank with the correct answer from the options given below:

- (a) 2.5×10^{-7} m (b) 4×10^{-7} m
(c) 5×10^{-7} m (d) 7.5×10^{-7} m

50. Radiation of frequency $2\nu_0$ is incident on a metal with threshold frequency ν_0 . The correct statement of the following is _____.

Fill in the blank with the correct answer from the options given below:

- (a) No photoelectrons will be emitted
(b) All photoelectrons emitted will have kinetic energy equal to $h\nu_0$
(c) Maximum kinetic energy of photoelectrons emitted can be $h\nu_0$
(d) Maximum kinetic energy of photoelectrons emitted will be $2h\nu_0$

Answer Key

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (a) | 2. (d) | 3. (c) | 4. (d) | 5. (b) | 6. (a) | 7. (a) | 8. (a) | 9. (c) | 10. (c) |
| 11. (b) | 12. (b) | 13. (d) | 14. (b) | 15. (a) | 16. (b) | 17. (d) | 18. (b) | 19. (b) | 20. (c) |
| 21. (b) | 22. (c) | 23. (a) | 24. (d) | 25. (d) | 26. (b) | 27. (b) | 28. (b) | 29. (d) | 30. (b) |
| 31. (a) | 32. (c) | 33. (d) | 34. (c) | 35. (c) | 36. (c) | 37. (a) | 38. (b) | 39. (d) | 40. (c) |
| 41. (b) | 42. (b) | 43. (a) | 44. (c) | 45. (b) | 46. (a) | 47. (a) | 48. (b) | 49. (c) | 50. (c) |

Explanations

- (a) For an electron in the ground state of a hydrogen atom:
– Kinetic Energy (KE) = (K)
– Potential Energy (PE) = $-2 \times \text{KE} = -2K$
– Total Energy (TE) = $\text{KE} + \text{PE} = K - 2K = -K$
- (d) The density of a nucleus is independent of its mass number. Therefore, the density ratio of any two nuclei is always 1 : 1.
- (c) As the point source causing photoelectric emission is moved away from the metallic plate, the intensity of light falling on the plate decreases with distance. This decrease in intensity reduces the photoelectric current.

The correct graph showing this inverse relationship between distance and photoelectric current is Graph 3.

- (d) The de Broglie wavelength (λ) of a particle is given by $\lambda = h/\sqrt{(2meV)}$, where h is Planck's constant, m is the mass, e is the charge, and V is the potential difference.

When the potential difference is doubled, the wavelength becomes $\lambda' = h/\sqrt{(2me(2V))} = \lambda/\sqrt{2}$.

Thus, the de Broglie wavelength decreases.

- (b) The series of the hydrogen spectrum are arranged by the shortest wavelengths as follows:

– Lyman series ($n = 1, n = \infty$) has the shortest wavelength.

– Balmer series ($n = 2, n = \infty$).

– Brackett series ($n = 4, n = \infty$).

– Pfund series ($n = 5, n = \infty$) has the longest wavelength.

The correct decreasing order is (A), (C), (B), (D).

- (a) For n-type semiconductors, elements with five valence electrons (pentavalent impurities) are used as dopants.

Arsenic (A) and Phosphorus (C) are pentavalent impurities, making them suitable for creating n-type semiconductors.

7. (a) The correct identification of the graphs is as follows:
 (A) represents a photo diode, as the current is proportional to light intensity and shows a reverse characteristic.
 (B) represents a solar cell, showing short-circuit current (I_{sc}) and open-circuit voltage (V_{oc}).
 (C) represents a Zener diode, with a sharp breakdown in reverse bias.
 (D) represents a forward biased p-n junction diode, showing exponential increase in current with voltage.
 The correct sequence is: (D), (C), (A), (B).
8. (a) The force on a current-carrying wire in a magnetic field is given by $F = I(L \times B)$, where L is the length of the wire in the direction of the field.
 For the straight segments force will be $2 \times BIR$ downward
 For the semicircular segment of radius R , the force will be directed downward and can be calculated as:
 $F = I(2R)B$ (since the effective length of the semicircle is $2R$)
 Thus, the force on the wire is $4BIR$, vertically downward.
9. (c) For an equilateral prism ($A = 60^\circ$), the angle of minimum deviation (δ_{min}) can be calculated using the formula for minimum deviation:
 $n = \sin[(A + \delta_{min})/2] / \sin(A/2)$
 Given $n = \sqrt{2}$ and $A = 60^\circ$:
 $\sqrt{2} = \sin[(60^\circ + \delta_{min})/2] / \sin(30^\circ)$
 $\sqrt{2} = \sin[(60^\circ + \delta_{min})/2] / 0.5$
 $\sin[(60^\circ + \delta_{min})/2] = 1/\sqrt{2}$
 $(60^\circ + \delta_{min})/2 = 45^\circ$
 $60^\circ + \delta_{min} = 90^\circ$
 $\delta_{min} = 30^\circ$
10. (c) The quantization of electric charge means that charge exists in discrete packets rather than in a continuous manner.
 This is evidenced by the transfer of integral numbers of electrons, as electrons are the fundamental charge carriers.
11. (b) The capacitance of a parallel plate capacitor with a dielectric is given by:
 $C = (\kappa \epsilon_0 A)/d$
 where κ is the dielectric constant, ϵ_0 is the permittivity of free space, A is the area of the plates, and d is the separation between the plates.
 Given that the distance has to be increased by 3.2 mm to restore the capacity, the effective distance with the dielectric slab is:

$$d_{eff} = d - t + t/\kappa$$

Where t is the thickness of the slab.

Substituting the given values, we get:

$$3.2 + 4/\kappa = 4 \quad [d - t = 3.2 \text{ \& } t = 4]$$

$$\kappa = 5$$

12. (b) Given the density of copper (ρ_c) = 8.0 g/cc = 8000 kg/m³ and the density of oil (ρ_o) = 0.8 g/cc = 800 kg/m³.

The volume of the ball (V) is given by:

$$V = (4/3)\pi r^3, \text{ where } r = 0.5 \text{ cm} = 0.005 \text{ m.}$$

$$V = (4/3)\pi(0.005)^3 = 5.24 \times 10^{-7} \text{ m}^3.$$

The mass of the ball (m) is:

$$m = \rho_c \times V = 8000 \times 5.24 \times 10^{-7} = 4.192 \times 10^{-3} \text{ kg.}$$

The buoyant force (F_b) is:

$$F_b = \rho_o \times V \times g = 800 \times 5.24 \times 10^{-7} \times 10 = 4.192 \times 10^{-3} \text{ N.}$$

The electric force (F_e) is:

$$F_e = qE, \text{ where } E = 600 \text{ N/C.}$$

To balance the forces

$$\rho E + 4/3\pi r^3 \times \rho_o \times g$$

$$= 4/3\pi r^3 \times \rho_{cu} \times g$$

$$\Rightarrow q = \frac{\frac{4}{3} \times 3.14 \times (0.005)^3 \times 7.2 \times 10^3 \times 10}{600}$$

$$= 6.288 \times 10^{-5} \text{ C}$$

no option match.

13. (d) When the temperature of a metal wire increases, the lattice ions vibrate more vigorously, which increases the rate of collisions between the electrons and the ions. This increased collision rate reduces the drift velocity (v_d) of the electrons. However, the thermal velocity (v_t) of the electrons increases with temperature.

Therefore, the drift velocity decreases and the thermal velocity increases.

14. (b) Let's simplify the given resistor network step by step:

1. Combine the two 4 Ω resistors in series at the top: $R_{s1} = 4 + 4 = 8 \Omega$.

2. Combine this 8 Ω with the 8 Ω resistor in parallel: $1/R_{p1} = 1/8 + 1/8 = 1/4$, so $R_{p1} = 4 \Omega$.

3. Combine the 4 Ω resistor with the 4 Ω resistors in series: $R_{s2} = 4 + 4 + 4 = 12 \Omega$.

4. Combine this 12 Ω with the 12 Ω resistor in parallel: $1/R_{p2} = 1/12 + 1/12 = 1/6$, so $R_{p2} = 6 \Omega$.

5. Now, combine the 6 Ω resistor with the 6 Ω resistors in series: $R_{s3} = 6 + 6 + 6 = 18 \Omega$.

Therefore, the total resistance between points A and B is 18 Ω .

15. (a) Given:

$$\text{Emf of first cell } (E_1) = 1.1 \text{ V}$$

$$\text{Internal resistance of first cell } (r_1) = 0.5 \Omega$$

$$\text{Resistance of wire } (R) = 0.5 \Omega$$

$$\text{Emf of second cell } (E_2) = 1.1 \text{ V}$$

Current remains the same when the second cell is connected in series.

Let the internal resistance of the second cell be r_2 .

The total emf when the second cell is connected in series is $E_1 + E_2 = 1.1 + 1.1 = 2.2 \text{ V}$.

The total internal resistance is $r_1 + r_2$.

Given that the current remains the same, the current through the circuit initially is:

$$I = E_1 / (r_1 + R)$$

$$I = 1.1 / (0.5 + 0.5)$$

$$I = 1.1 / 1$$

$$I = 1.1 \text{ A.}$$

When the second cell is connected in series, the current remains the same:

$$I = (E_1 + E_2) / (r_1 + r_2 + R)$$

$$1.1 = 2.2 / (0.5 + r_2 + 0.5)$$

$$1.1 = 2.2 / (1 + r_2)$$

$$1.1(1 + r_2) = 2.2$$

$$r_2 = 1 \Omega.$$

Therefore, the internal resistance of the second cell is 1 Ω .

16. (b) In a balanced Wheatstone bridge, the ratio of the resistances in one branch is equal to the ratio in the other branch:

$$P/Q = R/(S||x)$$

Where $S||x$ is the equivalent resistance of S and the shunt resistance x in parallel.

$$\text{Given:}, P = 3 \Omega$$

$$Q = 3 \Omega$$

$$R = 3 \Omega$$

$$S = 4 \Omega$$

The equivalent resistance of S and x in parallel is:

$$S||x = (S \times x) / (S + x)$$

For the bridge to be balanced:

$$3/3 = 3/(4||x)$$

$$1 = 3/(4||x)$$

$$4||x = 3 \Omega$$

Solving for x :

$$4||x = 3$$

$$4x / (4 + x) = 3$$

$$4x = 3(4 + x)$$

$$4x = 12 + 3x$$

$$x = 12 \Omega$$

Therefore, the resistance with which S must be shunted is 12 Ω .

17. (d) The magnetic moment (M) of a bar magnet is given by the product of the pole strength (m) and the length of the magnet (l):

$$M = m \times l$$

When the bar magnet is bent into a semicircular form, the length of the magnet becomes the semicircular length:

$$l = \pi r,$$

$$\text{So, } M = m \times \pi r$$

The new magnetic moment (M') is given by:

$$M' = m \times \ell'$$

Since the original length (l) of the bar magnet is $2r$ (where r is the radius of the semicircle):

$$M' = m \times 2r$$

$$\Rightarrow M' = \frac{2}{\pi} M$$

18. (b) Ferromagnetic materials used in transformers should have high permeability to allow the magnetic flux to pass through easily and to ensure efficient operation. Additionally, they should have low hysteresis loss to minimize energy losses during the magnetization and demagnetization cycles.

Therefore, the correct properties are high permeability and low hysteresis loss.

19. (b) The induced emf (ϵ) in a conducting ring is given by Faraday's Law of electromagnetic induction:

$$\epsilon = -d\Phi/dt$$

Where Φ is the magnetic flux.

For a ring of radius r , the magnetic flux Φ is given by:

$$\Phi = B \times A = B \times \pi r^2$$

The rate of change of magnetic flux ($d\Phi/dt$) is given by:

$$d\Phi/dt = \pi r^2 \times (dB/dt)$$

Given that the rate at which the magnetic field varies is x :

$$dB/dt = x$$

Therefore:

$$\epsilon = -\pi r^2 \times x$$

The electric field intensity (E) at any point of the ring is related to the induced emf by:

$$\epsilon = E \times 2\pi r$$

Thus:

$$E = -\pi r^2 \times x / 2\pi r$$

$$E = -rx / 2$$

Therefore, the electric field intensity at any point of the ring is $rx/2$.

20. (c) The voltage induced in the secondary of a transformer is given by the formula:

$$V_s = M \left(\frac{dI_p}{dt} \right)$$

Where V_s is the induced voltage, M is the mutual inductance, and $\frac{dI_p}{dt}$ is the rate of change of current in the primary.

For a sinusoidal AC current, the of current can be given by:

$$I_p = I_0 \times \sin(\omega t)$$

where I_0 is the crest value of the current, and ω is the angular frequency ($\omega = 2\pi f$).

The maximum rate of change of current $\left(\frac{dI_p}{dt} \right)$ is given by:

$$\frac{dI_p}{dt} = I_0 \times \omega \times \cos(\omega t)$$

The maximum value occurs when $\cos(\omega t) = 1$:

$$\frac{dI_p}{dt} = I_0 \times \omega$$

Substituting the given values:

$$I_0 = 1 \text{ A}$$

$$f = 50 \text{ Hz}$$

$$\omega = 2\pi \times 50 = 100\pi \text{ rad/s}$$

$$\frac{dI_p}{dt} = 1 \times 100\pi = 100\pi \text{ A/s}$$

Now, using the mutual inductance $M = 0.5 \text{ H}$:

$$V_s = 0.5 \times 100\pi = 50\pi \text{ V}$$

Approximating π to 3.14:

$$V_s \approx 50 \times 3.14 = 157 \text{ V}$$

The closest option is 150 V.

Therefore, the crest voltage induced in the secondary is 150 V.

21. (b) The induced emf (ϵ) in the coil is given by Faraday's law of electromagnetic induction:

$$\epsilon = -N \times (d\Phi/dt)$$

where N is the number of turns in the coil, and $d\Phi/dt$ is the rate of change of magnetic flux.

The magnetic flux (Φ) through the coil is given by:

$$\Phi = B \times A = \mu_0 n I \pi r^2$$

where μ_0 is the permeability of free space, n is the number of turns per unit length of the solenoid, I is the current, and r is the radius of the coil.

Given:

$$\mu_0 = 4\pi \times 10^{-7} \text{ Tm/A}$$

$$n = 2 \times 10^4 \text{ turns/m}$$

$$I = 4 \text{ A}$$

$$r = 0.01 \text{ m}$$

$$N = 100 \text{ turns}$$

The rate of change of current (dI/dt) is given by:

$$dI/dt = \Delta I / \Delta t = (4 - 0) / 0.05 = 80 \text{ A/s}$$

The rate of change of magnetic flux ($d\Phi/dt$) is:

$$d\Phi/dt = \mu_0 n \times (dI/dt) \times \pi r^2$$

$$d\Phi/dt = (4\pi \times 10^{-7}) \times (2 \times 10^4) \times 80 \times \pi \times (0.01)^2$$

Simplifying:

$$d\Phi/dt = 4\pi \times 10^{-7} \times 2 \times 10^4 \times 80 \times \pi \times 10^{-4}$$

$$d\Phi/dt = 4\pi \times 1.6\pi \times 10^{-3}$$

$$d\Phi/dt = 6.4\pi^2 \times 10^{-3} \text{ Wb/s}$$

The induced emf is:

$$\epsilon = -N \times (d\Phi/dt) = -100 \times 6.4\pi^2 \times 10^{-3} = -0.64\pi^2 \text{ V}$$

The total charge (Q) flowing through the coil is given by:

$$Q = \epsilon \times \Delta t / R$$

$$Q = -0.64\pi^2 \times 0.05 / (10\pi^2)$$

$$Q = -0.032\pi^2 / 10\pi^2$$

$$Q = -0.032 / 10 = -3.2 \times 10^{-3} \text{ C}$$

$$Q = 3.2 \times 10^{-3} \text{ C}$$

Converting to microcoulombs (μC):

$$Q = 3.2 \times 10^{-3} \times 10^6 = 32 \mu\text{C}$$

Therefore, the total charge flowing through the coil is 32 μC .

22. (c) When the lower half of a convex lens is made opaque, the lens still forms the complete image of the object. However, since only half of the lens is now effective in converging the light, the amount of light passing through the lens is reduced. This results in a decrease in the intensity of the image, but the full object is still visible.

(B) Image will show only half of the object – Incorrect.

(C) Intensity of image gets reduced – Correct.

(C) only – Intensity of image gets reduced

23. (a) The fringe separation (β) in a double-slit experiment is given by the formula:

$$\beta = (\lambda D) / d$$

where λ is the wavelength of the light, D is the distance between the slits and the screen, and d is the separation between the slits.

Given: $\lambda = 500 \text{ nm} = 500 \times 10^{-9} \text{ m}$

$D = 2 \text{ m}$

$d = 0.1 \text{ mm} = 0.1 \times 10^{-3} \text{ m}$

Substituting the values:

$$\beta = (500 \times 10^{-9} \times 2) / (0.1 \times 10^{-3})$$

$$\beta = (1000 \times 10^{-9}) / (0.1 \times 10^{-3})$$

$$\beta = (1000 / 0.1) \times 10^{-6}$$

$$\beta = 10^4 \times 10^{-6}$$

$$\beta = 10^{-2} \text{ m}$$

$$\beta = 1 \text{ cm}$$

Therefore, the fringe separation is 1 cm.

24. (d) The tube length (L) of an astronomical telescope is the sum of the focal lengths of the objective lens (f_o) and the eyepiece lens (f_e):

$$L = f_o + f_e$$

Given:

$$f_o = 10 \text{ m} = 1000 \text{ cm}$$

$$f_e = 10 \text{ cm}$$

Substituting the values:

$$L = 1000 + 10 = 1010 \text{ cm}$$

The magnification (M) of an astronomical telescope is given by the ratio of the focal length of the objective lens to the focal length of the eyepiece lens:

$$M = f_o / f_e$$

Substituting the values:

$$M = 1000 / 10 = 100$$

Therefore, the tube length and magnification of the telescope are 1010 cm and 100, respectively.

25. (d) According to Bohr's model:
- (A) The radius (r) of the orbiting electron is directly proportional to n^2 ($r \propto n^2$).
- (B) The speed (v) of the orbiting electron is directly proportional to $1/n$ ($v \propto 1/n$).
- (C) The magnitude of the total energy (E) of the orbiting electron is directly proportional to $1/n^2$ ($|E| \propto 1/n^2$).
- (D) The radius of the orbiting electron is not directly proportional to n.
- Therefore, the correct statements are: (B), (C) and (D).
26. (b) A full wave rectifier inverts every half cycle of the input AC signal, thereby doubling the frequency of the output signal.
- Therefore, if the input frequency is 50 Hz, the output frequency will be:
- Output frequency = $2 \times$ Input frequency = $2 \times 50 \text{ Hz} = 100 \text{ Hz}$
- Therefore, the output frequency of a full wave rectifier with an input frequency of 50 Hz is 100 Hz.

27. (b) For an electric dipole in a non-uniform electric field, the force and torque on the dipole depend on the orientation of the dipole moment (p) with respect to the electric field (E).

When the dipole moment is parallel to the direction of the field:

1. The force (F) on the dipole is non-zero ($F \neq 0$) because the field is non-uniform, causing a net force on the dipole.
2. The torque (τ) on the dipole is zero ($\tau = 0$) because the dipole is aligned with the field and there is no tendency for the dipole to rotate.

Therefore, the correct answer is $F \neq 0$ and $\tau = 0$.

28. (b) Consider the electric field (E) produced by the charged sheets:

1. At point P_1 , the electric field is due to both the sheets. So $F_1 = 0$.
2. At point P_2 , the electric fields due to both sheets are in same direction and don't get canceled.
3. At point P_3 , the electric field is zero due to both plates.

Therefore:

1. $F_1 = 0$ because the point charge experiences no force.
2. $F_2 \neq 0$ because the electric fields don't cancel out.
3. $F_3 = 0$ because the point charge experiences zero net field.

Answer: (2) $F_1 = 0$, $F_2 \neq 0$, $F_3 = 0$

29. (d) When two charged metallic spheres are brought into contact, they share charge until they reach the same potential. The potential (V) on the surface of a sphere is given by:

$$V = Q / R$$

where Q is the charge and R is the radius.

Since the potentials are equal when they are in contact:

$$V_1 = V_2$$

Therefore:

$$Q_1 / R_1 = Q_2 / R_2$$

Rearranging the equation to find the ratio of charges:

$$Q_1 / Q_2 = R_1 / R_2$$

Therefore, the ratio of the final charges on the two spheres is $Q_1/Q_2 = R_1/R_2$.

30. (b) According to Coulomb's Law, the force (F) between two point charges is given by:

$$F = k \times (q_1 \times q_2) / d^2$$

where k is Coulomb's constant, q_1 and q_2 are the charges, and d is the distance between them.

If each of the charges is doubled, the new force (F') will be:

$$F' = k (2q_1 \times 2q_2) / d'^2 = k \times (4q_1q_2) / d'^2$$

To keep the force unchanged ($F' = F$), we have:

$$k \times (4q_1q_2) / d'^2 = k \times (q_1q_2) / d^2$$

Simplifying, we get:

$$4 / d'^2 = 1 / d^2$$

$$d'^2 = 4d^2$$

$$d' = 2d$$

Therefore, the distance between the charges should be changed to 2d to keep the force unchanged.

31. (a) When capacitors are connected in series, the charge (Q) on each capacitor is the same. The voltage (V) across each capacitor is given by:

$$V = Q / C$$

Given two capacitors $C_1 = 2 \text{ } \mu\text{F}$ and $C_2 = 3 \text{ } \mu\text{F}$, the voltages V_1 and V_2 are:

$$V_1 = Q / C_1 = Q / 2$$

$$V_2 = Q / C_2 = Q / 3$$

The ratio of the voltages V_1 to V_2 is:

$$V_1 / V_2 = (Q / 2) / (Q / 3) = 3 / 2$$

The energy (U) stored in a capacitor is given by:

$$U = \frac{Q^2}{2C}$$

For capacitors in series, the energy stored in each capacitor is:

$$U_1 = \frac{Q^2}{2C_1}$$

$$U_2 = \frac{Q^2}{2C_2}$$

Substituting the values, we get:

$$U_1 = \frac{Q^2}{4}$$

$$U_2 = \frac{Q^2}{6}$$

The ratio of the energies U_1 to U_2 is:

$$\frac{U_1}{U_2} = \frac{6}{4} = \frac{3}{2}$$

32. (c) $200 \text{ } \Omega$ and $200 \text{ } \Omega$ are in parallel, so $R_2 = 100 \text{ } \Omega = r_{\text{eff}}$

The total resistance (R_{total}) of the circuit is:

$$R_{\text{total}} = R_1 + R_2 = 100 \text{ } \Omega + 100 \text{ } \Omega = 200 \text{ } \Omega$$

The current (I) in the circuit is given by Ohm's Law:

$I = V / R_{\text{total}} = 20 \text{ V} / 200 \Omega = 1/10 \text{ A}$
 The voltage (V) across the 200Ω resistor can be found using Ohm's Law:

$$V_{200} = IR_2 = 1/10 \text{ A} \times 200 \Omega = 10 \text{ V}$$

33. (d) For cells in parallel, the equivalent emf (E_{eq}) and internal resistance (r_{eq}) are calculated as follows:

Given:

$$E_1 = 2 \text{ V}, r_1 = 1 \Omega$$

$$E_2 = 1 \text{ V}, r_2 = 2 \Omega$$

The equivalent emf (E_{eq}) is given by:

$$E_{\text{eq}} = (E_1/r_1 + E_2/r_2) / (1/r_1 + 1/r_2)$$

Substituting the values, we get:

$$E_{\text{eq}} = (2/1 + 1/2) / (1/1 + 1/2) = (2 + 0.5) / (1 + 0.5) = 2.5 / 1.5 = 5/3 \text{ V}$$

The equivalent internal resistance (r_{eq}) is given by:

$$1/r_{\text{eq}} = 1/r_1 + 1/r_2$$

Substituting the values, we get:

$$1/r_{\text{eq}} = 1/1 + 1/2 = 1 + 0.5 = 1.5$$

$$r_{\text{eq}} = 1 / 1.5 = 2/3 \Omega$$

The total resistance in the circuit is the sum of the equivalent internal resistance and the external resistance ($R_{\text{ext}} = 4/3 \Omega$):

$$R_{\text{total}} = r_{\text{eq}} + R_{\text{ext}} = 2/3 + 4/3 = 6/3 = 2 \Omega$$

The current (I) in the circuit is given by Ohm's Law:

$$I = E_{\text{eq}} / R_{\text{total}} = (5/3) / 2 = 5/6 \text{ A}$$

Therefore, the current through the external resistance is $5/6 \text{ A}$.

34. (c) When a wire is stretched, its length increases and its cross-sectional area decreases. Given that the radius is halved, the area of cross-section (A) changes as follows:

$$A' = \pi(r/2)^2 = \pi r^2 / 4 = A / 4$$

The volume of the wire remains constant, so the new length (L') can be found using the original length (L):

$$A \times L = A' \times L'$$

$$A \times L = (A / 4) \times L'$$

$$L' = 4L$$

The resistance (R') of the stretched wire is given by:

$$R' = \rho \times (L' / A') = \rho \times (4L / (A / 4)) = \rho \times (16L / A) = 16 \times R$$

The resistivity (ρ') remains the same as resistivity is a material property:

$$\rho' = \rho$$

The power rating (P') at the same voltage (V) is given by:

$$P' = V^2 / R'$$

$$P' = V^2 / (16R) = (1/16) \times (V^2 / R) = (1/16) \times P$$

Therefore, the correct answer is $\rho' = \rho$, $R' = 16R$, $P' = (1/16)P$.

35. (c) Magnetic susceptibility (χ) measures how much a material will become magnetized in an applied magnetic field. The magnetic susceptibility of different types of materials is as follows:

1. Diamagnetics: These materials have a very small negative susceptibility. They are weakly repelled by a magnetic field. Examples include bismuth and copper.

2. Paramagnetics: These materials have a small positive susceptibility. They are weakly attracted to a magnetic field. Examples include aluminum and platinum.

3. Ferromagnetics: These materials have a very large positive susceptibility. They are strongly attracted to a magnetic field. Examples include iron, nickel, and cobalt.

Therefore, the order of materials in increasing order of magnetic susceptibility is:

Diamagnetics (B) < Paramagnetics (A) < Ferromagnetics (C).

36. (c) The force per unit length between two parallel conductors carrying currents I_1 and I_2 separated by a distance d is given by Ampère's force law:

$$F/L = (\mu_0 / 2\pi) \times (I_1 \times I_2) / d$$

where μ_0 is the permeability of free space.

Therefore, the total force F on a length L of one conductor due to the other is:

$$F = (\mu_0 / 2\pi) \times (I_1 \times I_2 \times L) / d$$

This shows that the force F is proportional to the product of the currents ($I_1 \times I_2$) and the length of the conductor (L).

Therefore, the correct answer is: proportional to $I_1 \times I_2 \times L$.

37. (a) To find the magnetic field at the center of the circular loop, we need to consider the contributions from the two semicircular parts separately.

The magnetic field at the center of a circular loop carrying current I is given by:

$$B = (\mu_0 I) / (2r)$$

For the semicircular part ABC:

$$B_{\text{ABC}} = \frac{1}{2} \times \frac{\mu_0 I}{2r} = \frac{\mu_0 I}{4r}$$

For the semicircular part ADC:

$$B_{\text{ADC}} = \frac{1}{2} \times \frac{\mu_0 2I}{2r} = \frac{2\mu_0 I}{4r}$$

The net magnetic field at the center will be the sum of these two fields. Since they are in the opposite direction:

$$B_{\text{total}} = B_{\text{ADC}} - B_{\text{ABC}}$$

$$= \frac{2\mu_0 I}{4r} - \frac{\mu_0 I}{4r} = \frac{\mu_0 I}{4r}$$

The direction of the magnetic field can be determined using the right-hand rule. For the given current direction, the field is out the plane.

38. (b) The torque (τ) experienced by a current-carrying loop in a magnetic field is given by:

$$\tau = n \times I \times A \times B \times \sin\theta$$

where:

n = number of turns (1 for a single loop)

I = current (10 A)

A = area of the loop ($1 \text{ cm}^2 = 1 \times 10^{-4} \text{ m}^2$)

B = magnetic field (0.2 T)

θ = angle between the normal to the plane of the loop and the magnetic field

Since the direction of the magnetic field is parallel to the plane of the loop, the angle θ between the normal to the plane of the loop and the magnetic field is 90 degrees ($\theta = 90^\circ$ or $\pi/2$ radians).

$$\sin 90^\circ = 1$$

Substituting the values, we get:

$$\tau = 1 \times 10 \text{ A} \times 1 \times 10^{-4} \text{ m}^2 \times 0.2 \text{ T} \times 1 = 2 \times 10^{-4} \text{ Nm}$$

Therefore, the torque experienced by the loop is $2 \times 10^{-4} \text{ Nm}$.

39. (d) In an AC circuit, the phase relationship between the current and the voltage depends on the nature of the circuit elements:

1. In a purely resistive circuit, the current and voltage are in phase (no phase difference).
2. In a purely inductive circuit, the current lags behind the voltage by $\pi/2$ (90 degrees).
3. In a purely capacitive circuit, the current leads the voltage by $\pi/2$ (90 degrees).

Given that the current leads the voltage by $\pi/2$, the circuit must be purely capacitive.

40. (c) Mutual inductance (M) between two coils is defined as the ratio of the change in magnetic flux ($\Delta\Phi$) to the change in current (ΔI) that causes it:

$$M = \Delta\Phi / \Delta I$$

Given:

$$\Delta\Phi = 15 \text{ Wb}$$

$$\Delta I = 10 \text{ A}$$

Substituting the values, we get:

$$M = 15 \text{ Wb} / 10 \text{ A} = 1.5 \text{ H}$$

Therefore, the mutual inductance of the coils is 1.5 H.

41. (b) For figure (a): The initial magnetic field is perpendicular into the plane. As the irregular wire turns into a circular loop, the area enclosed by the loop increases, increasing the flux into the plane. To oppose this increase, the induced current will flow in anticlockwise direction (producing a magnetic field out of the plane).

For figure (b): The initial magnetic field is perpendicular out of the plane. As the circular loop turns into a narrow straight wire, the area enclosed by the loop decreases, decreasing the flux out of the plane. To oppose this decrease, the induced current will flow in an anticlockwise direction.

Therefore, the correct answer is: clockwise in (a) and anticlockwise in (b).

42. (b)

1. Impedance (I): Impedance is a combination of resistance and reactance. It varies with frequency in complex circuits but generally doesn't show a straightforward linear or parabolic relationship as seen in specific reactances.
2. Capacitive reactance (II): Capacitive reactance (X_C) decreases with increasing frequency (f). The relationship is given by $X_C = 1 / (2\pi fC)$. This results in a hyperbolic curve decreasing with frequency.
3. Inductive reactance (III): Inductive reactance (X_L) increases linearly with increasing frequency. The relationship is given by $X_L = 2\pi fL$. This results in a straight line with a positive slope.
4. Resistance (IV): Resistance remains constant with frequency. It does not vary with frequency.

Therefore, the correct matching is:

(A) – (IV), (B) – (II), (C) – (III), (D) – (I)

43. (a) the ratio of energy densities of the electric and magnetic fields in an electromagnetic wave is 1 : 1.

44. (c) From the given options, the correct arrangement in decreasing order of wavelength is:

Radio waves, microwaves, Infrared waves, visible waves, X-rays.

45. (b) To match the electromagnetic waves with their production methods/devices, we need to understand how each type of wave is typically generated:

1. Microwaves: These are commonly produced using a device called a magnetron, which generates microwaves through the interaction of a stream of electrons with a magnetic field.
2. Infrared: Infrared waves are typically produced by the vibration and rotation of atoms and molecules.
3. X-rays: X-rays are produced by bombarding a large atomic number metal target with fast-moving electrons.
4. Radio waves: Radio waves are commonly generated using an LC (inductor-capacitor) oscillator.

46. (a) Given:

Radius of curvature (R) = 10 cm

Object distance (u) = -20 cm (since the object is on the left side of the surface)

Refractive index of material (μ) = 4/3

Using the formula for refraction at a spherical surface:

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

Here,

$\mu_1 = 1$ (refractive index of air)

– $\mu_2 = 4/3$ (refractive index of the material)

– v = image distance

Substituting the given values:

$$\frac{4/3}{v} - \frac{1}{-20} = \frac{4/3 - 1}{-10}$$

$$\Rightarrow v = -16 \text{ cm}$$

The negative value of v indicates that the image is formed to the left of the surface.

47. (a) The power (P) of a lens is given by the lens maker's formula:

$$P = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

μ is the refractive index of the material of the lens

(R_1) and (R_2) are the radii of curvature of the two surfaces of the lens

48. (b) The relationship between object distance (u), image distance (v), and focal length (f) for a convex lens is given by the lens formula:

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

This can be rearranged to find the image distance (v) as a function of object distance (u):

$$v = \frac{uf}{u - f}$$

The graph of this function is a hyperbola. It will be in the 2nd Quadrant as u is negative.

49. (c) Given:

Width of slit (a) = 0.1 mm = 0.1×10^{-3} m

Linear width of central maxima (β) = 5 mm = 5×10^{-3} m

Distance between slit and screen (D) = 50 cm = 0.5 m

The width of the central maxima in a single slit diffraction pattern is given by:

$$\beta = \frac{2\lambda D}{a}$$

$$\lambda = \frac{\beta a}{2D}$$

Substituting the given values:

$$\lambda = \frac{(5 \times 10^{-3}) \times (0.1 \times 10^{-3})}{2 \times 0.5}$$

$$\lambda = \frac{5 \times 10^{-4}}{1}$$

$$\lambda = 5 \times 10^{-7} \text{ m}$$

50. (c) According to the photoelectric effect, the energy of the incident photons is given by:

$$E = h(2\nu_0)$$

The threshold energy required to emit photoelectrons is given by:

$$E_0 = h\nu_0$$

The excess energy is converted into the kinetic energy of the emitted photoelectrons. Therefore, the maximum kinetic energy (K.E) of the emitted photoelectrons is given by:

$$\text{K.E} = h(2\nu_0) - h\nu_0$$

$$\text{K.E} = h\nu_0$$